

An Analytical Model of Critical and Subcritical Alkali Metal Dendrite Growth in Ceramic Solid Electrolytes

Ansgar Lowack

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Abstract

In solid-state batteries, ceramic solid electrolytes are penetrated by dendrites when plating above a critical current density J_{crit} . A dendrite will propagate by metal deposition at a pre-existing dendrite tip if the mechanical energy required to crack the ceramic open is less than the electrical energy (Joule heating) wasted by forcing the current to detour around the dendrite to the flat electrode surface. Based on this principle of minimal power dissipation, a dependence of $J_{\text{crit}} \propto c_{\text{max}}^{3/2}$ is derived. c_{max} is the length of the longest preexisting, sufficiently thin interfacial defect. Furthermore, the theory is expanded to include electrochemical stress-corrosion-cracking at dendrite tips due to residual electron conduction of the solid electrolyte. The resulting subcritical dendrite growth follows the same defect dependence. Consequentially, scattering of dendrite growth between samples must follow a Weibull-distribution, similar to the tensile strength of ceramic components but at smaller Weibull-modulus.

1 Introduction

Ceramic solid electrolytes are often considered promising candidates for enabling the use of lithium or sodium electrodes in next generation high energy-density batteries for room-temperature applications [1, 2, 3]. In contrast to conventional liquid electrolytes, it is intuitive to believe that separators made from a dense ceramic solid electrolytes are a mechanical barrier which prevents dendrite penetration. However, this intuition has turned out to be wrong as experiments on a wide range of materials have demonstrated dendrite penetration through ceramic solid electrolytes.

The mechanism by which penetration happens is still under active investigation. A growing body of theoretical [4, 5, 6, 7, 8, 9] and experimental [10, 11, 12] work presents strong evidence that crack growth within the solid electrolyte is the dominant pathway for metal penetration. Assuming this hypothesis is correct, there is a notable lack of analytical expressions linking the critical current density in ceramic ion conductors to defect sizes, thus providing intuitive, design-oriented guidance for dendrite-resistant solid electrolytes.

The present work addresses this gap by deriving an analytical formula for the critical current density in ceramic solid electrolytes. Based on the semi-empirical understanding previously presented in [12]. An analytical expression for the reduction in Joule dissipation associated with alkali metal deposition at the tip of an interfacial defect is derived. To achieve this, interfacial defects are approximated as flat ellipsoids with symmetry points on the interfacial plane between electrode and solid electrolyte. This approximation turns finding the electrical potential (and hence the Joule dissipation) around the defect from only numerically solvable into a classical electrodynamics problem with known analytical solution. From Griffith's theory of critical crack-growth, an analytical expression for the mechanical dissipation associated with alkali metal deposition at the tip of this defect and consequential crack propagation is calculated. Minimizing total dissipation as the sum of Joule and mechanical dissipation yields the current density at which thermodynamics enable critical crack growth driven dendrite propagation. In addition to this picture of electrochemically induced critical crack growth, residual electron conduction of the solid electrolyte is analytically coupled to

the system failure. The resulting theory predicts subcritical stress-corrosion-cracking at defect tips during alkali metal deposition below the critical current density. By the suggested mechanism dendrites grow, on a timescale which can be suppressed by decreasing the residual electron conductivity, until critical failure onset. Since both critical and subcritical dendrite growth depend on the size of interfacial defects, a weakest link behavior is postulated. As a consequence it is shown that dendrite formation will scatter between samples of similar preparation following a Weibull-distribution.

2 Discussing the model

2.1 Geometrical assumptions

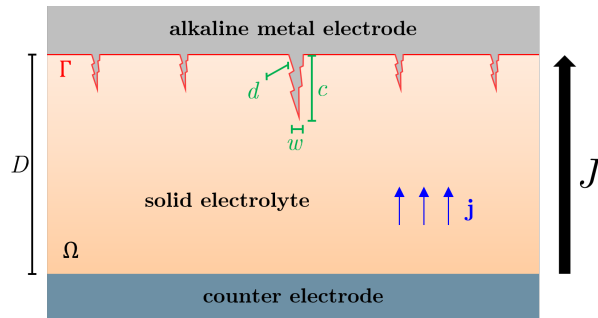


Figure 1: Schematic cross section of a SSB with unspecified counter electrode, indicating relevant variables.

A schematic sketch of the solid-state battery is depicted in **Figure 1**. Ω is the volume of the solid electrolyte with thickness D and cross-sectional area A_{cell} . Let Γ be the interface to the alkali metal electrode with surface area A_{Γ} . The solid electrolyte is assumed to be homogeneous and isotropic. Microstructural defects within the bulk, such as pores and grain boundaries, are neglected. This simplification is supported by prior experimental observations that dendrites nucleate at the interface and propagate in a stable manner once the critical current density is exceeded [12]. Hence, bulk defects in the solid electrolyte are not considered to play a determining role in either the nucleation mechanism or the critical current density value.

In contrast, interfacial defects act as preferential nucleation sites for dendrite formation. These defects are modeled as a distribution of N cracks located at positions $\mathbf{x}$ along the interface Γ , described by characteristic penetration length scales $c(\mathbf{x})$, characteristic width scales $w(\mathbf{x})$ and characteristic depth scales d . Depending on the material chemistry, grain sizes, and processing conditions, c is bounded below by a global minimum value $c_{\min}$ and is assumed to follow a Pareto distribution, yielding:

$$P(C > c) = \left(\frac{c_{\min}}{c} \right)^k \quad \text{for} \quad D \gg c \geq c_{\min}, \quad k > 0. \quad (1)$$

k is the shape parameter (also frequently referred to as the Pareto index or tail index), i.e. a measure of how diverse the crack sizes are in the electrolyte ceramic: Small k (e.g., $k \approx 1$) mean there is a high probability of finding very large cracks compared to the average. Large k (e.g., $k > 10$) mean the cracks are very uniform in size and the material is highly consistent.

The defects are assumed to be filled with the alkali metal, either due to alkali metal creep during electrode contacting or prior alkali metal deposition at subcritical current density.

2.2 Transport

The ion transport in the solid electrolyte is described by the vector field of current density $\mathbf{j}(\mathbf{x}, J)$ which is source-free and follows from Ohm's law in Ω with ionic conductivity σ , electric field $\mathbf{E}(\mathbf{x}, J)$ and electrical potential $\Phi(\mathbf{x}, J)$:

$$\nabla \cdot \mathbf{j}(\mathbf{x}, J) = 0, \quad \mathbf{j}(\mathbf{x}, J) = \sigma \mathbf{E}(\mathbf{x}, J) = \sigma \nabla \Phi(\mathbf{x}, J), \quad \mathbf{x} \in \Omega. \quad (2)$$

At the interface, $\mathbf{j}(\mathbf{x}, J)$ is subject to the global constraint

$$J = \frac{1}{A_{\text{cell}}} \int_{\Gamma} j_n(\mathbf{x}, J) \, dA, \quad j_n(\mathbf{x}, J) = \mathbf{j}(\mathbf{x}, J) \cdot \mathbf{n}(\mathbf{x}), \quad \mathbf{x} \in \Gamma, \quad (3)$$

where $\mathbf{n}(\mathbf{x})$ is the unit normal vector of Γ at $\mathbf{x}$ and J is the global current density controlled by the experimentalist.

2.2.1 Dissipation

Bulk Joule dissipation: The current field $\mathbf{j}$ is associated with power dissipation in the solid electrolyte via Joule heating:

$$\dot{Q}_{\text{bulk}}(J) = \int_{\Omega} \rho |\mathbf{j}(\mathbf{x}, J)|^2 \, dV, \quad \rho = \sigma^{-1}. \quad (4)$$

Interfacial Butler-Volmer dissipation: To deposit the alkali metal at the electrode, interfacial reaction kinetics lead to additional power dissipation, as described by the Butler-Volmer equation. This effect is neglected in this model, since it is known to be very small between alkali metal electrodes and clean oxide ceramic solid electrolytes which are sufficiently stable (and thus technologically relevant). For example Krauskopf et al. [13] experimentally concluded the interface resistance between lithium and LLZO to be practically $0 \, \Omega \text{cm}^2$ at high pressure. It is evident that this high-pressure condition is met at the tip of dendrites, which continuously propagate as cracks in the ceramic. Non-linear effects would only decrease the effect further, due to the concave relation between local interfacial overpotential and current density described by the Butler-Volmer equation.

Interfacial mechanical dissipation: Once the ion has crossed the interface, it must be incorporated into the lattice of the metal electrode, leading to an area-specific local volume injection rate $\dot{v}(j_n(\mathbf{x}, J))$:

$$\dot{v}(j_n(\mathbf{x}, J)) = \frac{V_M}{F} j_n(\mathbf{x}, J), \quad \mathbf{x} \in \Gamma, \quad (5)$$

with molar volume V_M of the alkali metal. Under the assumption of complete geometric contact between solid electrolyte and electrode, this volume injection dissipates mechanical power

$$\dot{W}(j_n(\mathbf{x}, J)) = \int_{\Gamma} \dot{v}(j_n(\mathbf{x}, J)) p(\mathbf{x}, J) \, dA = \int_{\Gamma} \frac{V_M}{F} p(\mathbf{x}, J) j_n(\mathbf{x}, J) \, dA, \quad (6)$$

with an interfacial pressure field $p(\mathbf{x}, J)$. Under specific circumstances this pressure field drives the dendrite formation, as discussed in the next section

2.2.2 Interfacial pressure field

Clearly, the value of the pressure field p in equation 6 differs significantly between sites with and without defect.

Volume injection at site with no defect: To deposit alkali metal at a site $\mathbf{x}_0 \in \Gamma$ on the interface where there is no defect, the volume injection $\dot{V}(j_n(\mathbf{x}_0))$ must be accommodated. This can be achieved by

stress-driven creep in the alkali metal electrode. Generally, due to the ductile nature of lithium and sodium at room temperature, the pressure required for this is small compared to the pressure required to propagate a crack in the ceramic solid electrolyte. Thus, without any defect:

$$p(j_n(\mathbf{x}_0)) =: p_{\text{creep}}(j_n(\mathbf{x}_0)) \approx 0. \quad (7)$$

Volume injection at a defect: The situation changes at some $\mathbf{x}_c \in \Gamma$, where there is a defect of length c , width w and depth d . As depicted in **Figure 2**, now two modes of volume generation at the tip are possible.

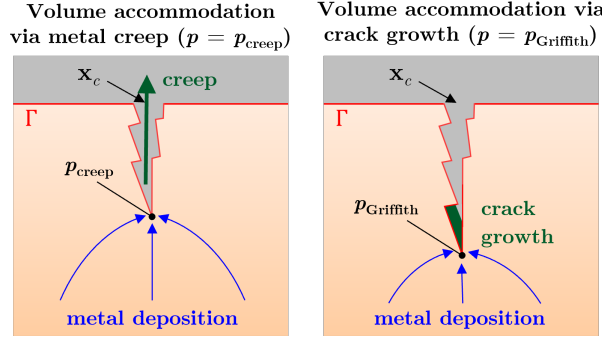


Figure 2: Schematic depiction of the alkali metal volume injection in a crack-like interfacial defect via metal creep (left) and via crack growth (right).

If $\dot{v}(j_n(\mathbf{x}_c))$ is small enough, again a pressure $p(j_n(\mathbf{x}_c)) = p_{\text{creep}}(j_n(\mathbf{x}_c))$ will build up until the creep rate in the alkali metal is large enough for volume accommodation (Figure 2, left). However, if the defect is mechanically confined, $p_{\text{creep}}(j_n(\mathbf{x}_c))$ will increase to large values. An accurate description of the metal creep in a confined micro-crack is theoretically challenging, due to the non Newtonian behavior of alkali metal deformation under pressure and nano-scale effects. However, a rough estimation can be made by approximating the cross-section of the defect as a slid of width $w \ll d$. In this geometry, the alkali metal creep above some yield stress is approximated as viscoplastic flow in a pressure gradient $\nabla p = p/c$. Following the derivation shown e.g. by Shapovalov [14], this yields the total volume creep $\dot{V}(j_n(\mathbf{x}_c, J))$ through the defect as:

$$\dot{V} = w^2 d \frac{2n}{1+2n} \left(\frac{p w}{G c} \right)^{1/n} \quad (8)$$

Here, G and n are defined by the relation between stress τ and strain rate $\dot{\gamma}$ via the Norton power law:

$$\tau = G \dot{\gamma}^n \quad (9)$$

Normalizing $\dot{V}$ to the cross-section $w \cdot d$ of the slid yields:

$$\dot{v}(j_n(\mathbf{x}_c, J)) = \frac{\dot{V}(j_n(\mathbf{x}_c, J))}{w d} = w \frac{2n}{1+2n} \left(\frac{p_{\text{creep}}(j_n(\mathbf{x}_c)) w}{G c} \right)^{1/n} \quad (10)$$

Rearranging and insertion of equation 5 yields:

$$p_{\text{creep}}(j_n(\mathbf{x}_c, J)) = \left(\frac{V_M}{F} \frac{1+2n}{2n} j_n(\mathbf{x}, J) \right)^n \frac{G c}{w^{1+n}} \quad (11)$$

This shows that $p_{\text{creep}}(j_n(\mathbf{x}_c, J))$ grows quickly with $j_n(\mathbf{x}_c, J)$ for defects where $w \ll c$.

However, pressure inside such a crack cannot grow indefinitely with $j_n(\mathbf{x}_c, J)$. At some pressure the crack will start to propagate into the solid electrolyte, creating volume at its tip which can be filled with the depositing

alkali metal. This mechanism is depicted on the right hand side in in Figure 2. The pressure inside the defect, required to propagate the crack-like defect (opening mode I), is given by Griffith's equation:

$$p_{\text{Griffith}}(\mathbf{x}_c) = \frac{K_{Ic}}{Y_0 \sqrt{c}} \quad (12)$$

where K_{Ic} is the fracture toughness of the solid electrolyte and $Y_0 \sim 1$ is a geometrical constant which shall be omitted in the following. Notice that in the context of figure 2, K_{Ic} describes the fracture toughness for the alkali metal filled crack. It is smaller than the fracture toughness for a dry crack K_{Ic}^{dry} , as measured in e.g. tensile strength tests in air. Using the Griffith-Irwin-equation and Dupré's equation, the reduction follows from the surface energy γ_{ceramic} of the solid electrolyte, the surface energy γ_{metal} of the alkali metal and the interfacial energy $\gamma_{\text{ceramic/metal}}$ as well as the work of adhesion W_{adh} between both materials:

$$K_{Ic} = K_{Ic}^{\text{dry}} \sqrt{\frac{\gamma_{\text{ceramic/metal}}}{\gamma_{\text{ceramic}}}} = K_{Ic}^{\text{dry}} \sqrt{\frac{\gamma_{\text{ceramic}} + \gamma_{\text{metal}} - W_{\text{adh}}}{\gamma_{\text{ceramic}}}} \quad (13)$$

Consider now that $p_{\text{creep}}(j_n(\mathbf{x}_c))$ can only increase with current until some value $j_{\text{Griffith}}(\mathbf{x}_c)$, where

$$p_{\text{creep}}(j_{\text{Griffith}}(\mathbf{x}_c)) = p_{\text{Griffith}}(\mathbf{x}_c). \quad (14)$$

Insertion of equation 11 and 12 then yields:

$$j_{\text{Griffith}}(\mathbf{x}_c) \approx \frac{2n}{1+2n} \frac{F}{V_M} \left(\frac{K_{Ic}}{G c^{3/2}} \right)^{1/n} w^{(1+n)/n} \quad (15)$$

This value describes the current density at or above which alkali metal must be deposited at the defect tip to thermodynamically allow crack growth. The effective area of the defect tip, where the alkali metal is deposited, scales for small w like $d \cdot c$. Thus, as a necessary (but not sufficient) condition: For a dendrite to grow at $\mathbf{x}_c$, a current

$$I_c \gtrsim d c j_{\text{Griffith}}(\mathbf{x}_c) = \frac{2n}{1+2n} \frac{F}{V_M} \left(\frac{K_{Ic}}{G} \right)^{1/n} w^{(1+n)/n} d c^{1-3/2n} \quad (16)$$

must be deposited at the tip of a defect, to propagate the crack. I_c decreases rapidly in value with decreasing w . Thus, only sufficiently thin defects with a low width w will be dangerous in the context of dendrite formation.

2.3 Onsager's principle of minimal dissipation

At a given global current density J , of all current fields $\mathbf{j}$ satisfying equation 2 to 6 mathematically, the physically realized one minimizes the total irreversible power dissipation. This variational principle is (in this specific case) a direct consequence of the second law of thermodynamics and corresponds to the colloquial term "The electric current takes the path of lowest resistance".

When neglecting all other minor contributions, the dissipation functional which assigns the total dissipated power to a current field $\mathbf{j}$ at J follows from equation 4 and 6 as:

$$\mathcal{D}[\mathbf{j}] = \int_{\Omega} \rho |\mathbf{j}(\mathbf{x}, J)|^2 dV + \int_{\Gamma} \frac{V_M}{F} p(\mathbf{x}, J) j_n(\mathbf{x}, J) dA. \quad (17)$$

2.4 Emergence of a local critical current density

There is only one dendrite across the whole solid electrolyte necessary to cause an irreversible failure of the cell. The local critical current density J_{crit}^L of this growing dendrite ultimately determines the observed

overall critical current density. According to equation 16 this dendrite most probably originates from a thin and well above average large interfacial defect – i.e. an ‘outlier’ from the high- c tail of the Pareto distribution in equation 1. Assume that these outliers, i.e. potentially dangerous defects, are spaced far enough apart to act independently from each other. This allows to analyze the failure of the cell by looking at how one single, isolated crack interacts with the current, without worrying about other cracks nearby. Thus, a correlation volume $\Omega_c \subset \Omega$ around each of the low w/c ratio defects of length c at $\mathbf{x}_c$. Within this region, the electrical potential is governed solely by that specific defect. Outside of this volume, the defect has no impact, allowing the electrical potential Φ and current density $\mathbf{j}$ to return to their uniform, bulk states:

$$\Phi(\mathbf{x}) =: \begin{cases} \Phi_c(\mathbf{x}) & \text{for } \mathbf{x} \in \Omega_c \\ \Phi_0 \approx -E_0 z & \text{for } \mathbf{x} \notin \Omega_c \end{cases} \Rightarrow \mathbf{j}(\mathbf{x}) =: \begin{cases} \mathbf{j}_c(\mathbf{x}) = -\sigma \nabla \Phi_c(\mathbf{x}) & \text{for } \mathbf{x} \in \Omega_c \\ \mathbf{j}_0(\mathbf{x}) \approx \sigma E_0 \mathbf{e}_z & \text{for } \mathbf{x} \notin \Omega_c \end{cases}, \quad (18)$$

where $\sigma E_0 = J$ is externally fixed, z is the vertical component of $\mathbf{x}$ and $\mathbf{e}_z$ the unit vector in z -direction. This is a purely statistical assumption which requires k in equation 1 to be sufficiently small. For ceramic solid electrolytes this can be safely assumed as their processing is still in development - in contrast to e.g. high-end structural ceramics like ZrO_2 where defect scattering has been reduced by decades of engineering progress. The integral constraint of equation 3 now allows for different shapes of $\Phi_c(\mathbf{x})$ and $\mathbf{j}_c(\mathbf{x})$ in Ω_c . The two extreme cases are illustrated in **Figure 3** and will be elaborated in the following.

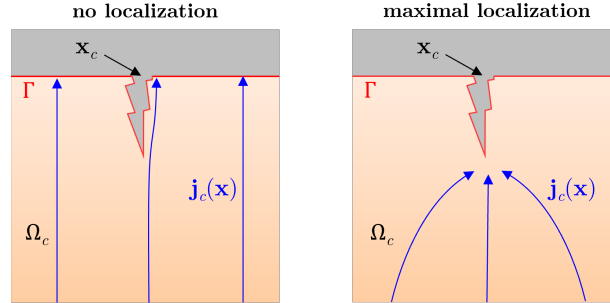


Figure 3: Schematic visualization of the current field $\mathbf{j}_c(\mathbf{x})$ around a defect at $\mathbf{x}_c$ for no current localization and the maximum possible current localization.

No current localization: If the mechanical dissipation term for metal deposition at the defect is considered infinitely large compared to the Joule dissipation term, deposition at the defect and hence current localization never minimizes the dissipation functional (equation 17). This is always the case for sufficiently small values of J , since the Joule dissipation increases faster with J than the mechanical dissipation. Here, J is locally subcritical, $J < J_{\text{crit}}^L$. The shape of the current field is homogeneous, except for the current which must be redistributed from the volume of the field-free metal-filled defect. This is sketched in Figure 3 on the left hand side. If the defect is sufficiently long and thin, the dissipation change due to this perturbation is small compared to the maximal localized case. Thus, the potential and current field are approximated as the trivial solution $\Phi_c(\mathbf{x}) = \Phi_0$ and $\mathbf{j}_c(\mathbf{x}) = \mathbf{j}_0$.

Maximal current localization: The right hand side of Figure 3 illustrates the maximal localized current field $\mathbf{j}_c$ around the alkali metal filled defect - i.e. the emerging dendrite - which is *electro-dynamically* allowed. The field follows from the solution Φ_c of the Laplace equation for the depicted geometry, when the whole electrolyte/metal interface (including the dendrite) is considered to be at constant electric potential as a Dirichlet boundary condition. Consequently, it represents the current field of minimal Joule dissipation. Compared to the case of no current localization, Joule dissipation is reduced by:

$$\Delta \dot{Q}_{\text{max}}(J) = \sigma \int_{\Omega_c} |\nabla \Phi_c|^2 dV - \sigma \int_{\Omega_c \cup V_c} E_0^2 dV \quad (19)$$

Here, V_c is the volume inside the defect, which is filled with alkali metal. Equation 19 is the maximum potential dissipation reduction by current localization. For current localization to minimize overall dissi-

pation and thus equation 17, the corresponding mechanical dissipation cost $\Delta\dot{W}(J)$ cannot be larger than $\Delta\dot{Q}_{\max}(J)$. Since both terms in equation 17 increase with J and Joule dissipation scales quadratic, while mechanical dissipation scales linearly, dendrite growth at $\mathbf{x}_c$ is thermodynamically possible (i.e. minimizes total dissipation in Ω_c) as soon as

$$\Delta\dot{Q}_{\max}(J = J_{\text{crit}}^L) + \Delta\dot{W}(J = J_{\text{crit}}^L) = 0, \quad (20)$$

assuming equation 16 is fulfilled at $\mathbf{x}_c$. This defines the local critical current density $J_{\text{crit}}^L(\mathbf{x}_c)$.

Intuition of a local critical current density: Consider the defect as a crack in the ceramic of characteristic width w , characteristic depth d and characteristic length c (Figure 1), e.g. between two grains, which comes to some sort of point at its tip. Under maximal current localization (Figure 3, right) it is intuitively clear that the current density will focus on this tip of the defect. This produces a volume of locally higher Joule dissipation which should scale like the volume of the defect itself, thus like $w \cdot d \cdot c$. At the foot of the defect, closer to the electrode, the current density will be decreased, resulting in a volume of reduced Joule dissipation which will also scale like $w \cdot d \cdot c$. At the base, an area scaling as $w \cdot d$ is effectively shadowed from current, as the corresponding current $I_c \propto J w \cdot d$ is deposited at the tip of the defect. Following equation 6, 12 and 20, and considering Joule dissipation scales like J^2 , a reasonable intuition is therefore:

$$\Delta\dot{W}(J) \propto J p_{\text{Griffith}} w d \propto J \frac{w d}{\sqrt{c}}, \quad \Delta\dot{Q}_{\max}(J) \propto J^2 w d c \quad \Rightarrow \quad J_{\text{crit}}^L \propto c^{-3/2} \quad (21)$$

To underline this intuition mathematically for an analytically solvable example and estimate proportionality constants, the intuitively described defect is now modeled as one half of an ellipsoid. This geometry is flexible enough to believably describe a thin crack in a ceramic. Both $\Delta\dot{Q}_{\max}$ and $\Delta\dot{W}$ must be estimated using this approximation.

2.5 Dissipation estimation

Estimating Joule dissipation: The Laplace equation must be solved to determine Φ_c and calculate the dissipation field. For an arbitrarily shaped defect, this problem has no analytical solution. However, if the defect is approximated to be one half of an ellipsoid the problem can be treated analytically using the method of image charges. For rigorous treatment of this problem, section 3.26 of Stratton's book "Electromagnetic Theory" [15] or section 4 of Landau & Lifschitz's book "Electrodynamics of Continuous Media" [16] is recommended. The resulting geometrical assumptions are schematically illustrated in **Figure 4**.

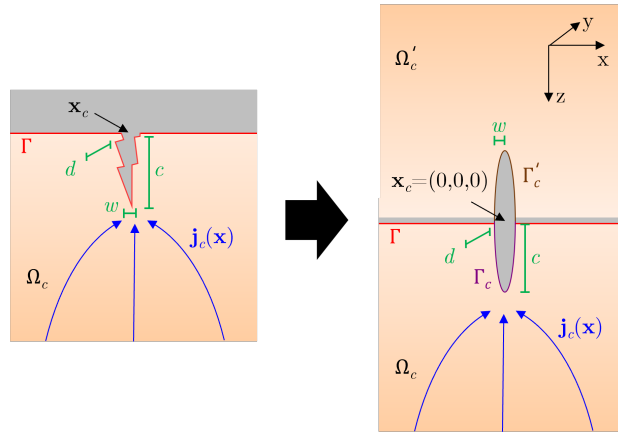


Figure 4: Approximation of defect as an ellipsoid using the method of mirror charge. The defect surface is Γ_c . Hence, the surface of the full ellipsoid is $\Gamma_c \cup \Gamma'_c$.

The problem is parameterized in confocal ellipsoidal coordinates (λ, μ, ν) which follow from the equation

$$\frac{x^2}{w^2 + u} + \frac{y^2}{d^2 + u} + \frac{z^2}{c^2 + u} = 1, \quad (22)$$

as

$$\lambda \in (-c^2, \infty), \quad \mu \in (-d^2, -c^2), \quad \nu \in (-w^2, d^2). \quad (23)$$

In this parametrization, $\lambda = 0$ describes the surface of an ellipsoid with semi-axis w, d, c , centered at $(0, 0, 0)$. This surface is defined as $\Gamma_c \cup \Gamma'_c$ in Figure 4. The following shorthand is defined to enhance readability:

$$R(s) = \sqrt{(w^2 + s)(d^2 + s)(c^2 + s)} \quad (24)$$

Following Stratton (§3.26, equation (26) in [15]) or Landau & Lifschitz (§4, equation (4.24) in [16]), the exterior electrostatic potential of a conducting ellipsoid held at zero potential in a uniform field $E_0 \mathbf{e}_z$ is

$$\Phi(\lambda, z) = -E_0 z \left[1 - \frac{\mathcal{I}(\lambda)}{\mathcal{I}(0)} \right], \quad (25)$$

with the ellipsoidal integral:

$$\mathcal{I}(\lambda) = \frac{w d c}{2} \int_{\lambda}^{\infty} \frac{ds}{(s + c^2) R(s)} \quad (26)$$

The interface between electrode and solid electrolyte is introduced as a mirror plane through the ellipsoid at $z = 0$. By this method of image charges, the potential given by Stratton (equation 25) is the potential $\Phi(\lambda, z > 0) = \Phi_c(\lambda, z)$ in the volume Ω_c inside the solid electrolyte around the ellipsoid, defined only on the lower half-space $z > 0$. Consequently, the surface of the defect is Γ_c in Figure 4.

Equation 25 is now inserted into equation 19. This yields after some calculation (shown in section 4.2):

$$\Delta \dot{Q}_{\max}(J) \approx -\frac{2\pi \sigma E_0^2 w d c}{3 \mathcal{I}(0)} = -\frac{2\pi J^2 w d c}{3 \sigma \mathcal{I}(0)} \quad (27)$$

Thus confirming the intuition (equation 21). The geometric shape parameter enters through $\mathcal{I}(0)$ (equation 26) which has no closed-form expression.

Mechanical dissipation: The total current I_c which must be deposited at the defect to realize the reduction of Joule dissipation in equation 19 is

$$I_c = \int_{\Gamma_c} \mathbf{j}_c \cdot \mathbf{n} dA = -\sigma \int_{\Gamma_c} \nabla \Phi_c \cdot \mathbf{n} dA, \quad (28)$$

where $\mathbf{n}$ is the outward unit normal on the ellipsoid surface Γ_c . The computation of this integral after insertion of the ellipsoid potential (equation 25), with the calculation shown in section 4.3, yields:

$$I_c \approx \frac{\pi \sigma E_0 w d}{\mathcal{I}(0)} = \frac{\pi J w d}{\mathcal{I}(0)} \quad (29)$$

If the defect is sufficiently thin, I_c will always fulfill equation 16 and thus the mechanical dissipation change associated with current localization follows from equation 6 and 12 as

$$\Delta \dot{W}(J) = \frac{V_M}{F} p_{\text{Griffith}}(\mathbf{x}_c) I_c \approx \frac{\pi V_M K_{Ic} J w d}{F \mathcal{I}(0) \sqrt{c}}. \quad (30)$$

Again, the intuition (equation 21) is confirmed. The geometric shape parameter enters through $\mathcal{I}(0)$ (equation 26).

2.6 Global critical current density

The estimations in equation 27 and 30 can now be inserted into the local critical current criterion of equation 20. The integral $\mathcal{I}(0)$ cancels out and rearranging yields the local critical current density:

$$J_{\text{crit}}^L(\mathbf{x}_c) \approx \frac{3 \sigma V_M K_{Ic}}{2 F c^{3/2}} \Rightarrow J_{\text{crit}}^L \approx \frac{Y \sigma V_M K_{Ic}}{F c^{3/2}}, \quad (31)$$

In this derivation, the geometry constant of the intuitive derivation in equation 21 is $3/2$. Reasonably, for differently shaped defects which are not perfectly describable as an ellipsoid, the functional dependence should remain approximately the same, but the numerical prefactor might deviate. Hence, it is substituted by a geometric constant $Y \sim 3/2$. As it turns out, equation 31 follows as a limit of the more advanced numerical model of Mukherjee et al. with different boundary conditions [17].

Equation 31 applies for all $\mathbf{x}_c \in \Gamma$ with low width w defects of length c . The weakest link of Γ - i.e. the dendrite which will first penetrate the solid electrolyte - is the defect with the longest length $c = c_{\text{max}}$. The global critical current density thus follows as:

$$J_{\text{crit}} \approx \frac{Y \sigma V_M K_{Ic}}{F c_{\text{max}}^{3/2}} \quad (32)$$

In the rough approximation of viscoplastic flow, the term "sufficiently thin" can be specified by insertion of equation 29 in equation 16. The defect of length c_{max} in equation 32 must have a sufficiently small width w to fulfill:

$$J_{\text{crit}} \gtrsim \frac{2n}{1+2n} \frac{F \mathcal{I}(0)}{\pi V_M} \left(\frac{K_{Ic}}{G} \right)^{1/n} c_{\text{max}}^{1-3/2n} w^{1/n} \quad (33)$$

Here, the dependence on $\mathcal{I}(0)$ remains. As Landau & Lifschitz demonstrate (§ 4, equation (4.29) in [16]) $\mathcal{I}(0) \leq 1$, depending on the shape of the ellipsoid (and independent of its volume). For $w = d \ll c$:

$$\mathcal{I}(0) = \frac{w^2}{c^2} \ln \left(\frac{c}{w} \right) \quad (34)$$

2.7 Subcritical dendrite formation

Equation 32 describes critical crack growth for sufficiently fast metal deposition. This is however not the only plausible failure mode. In principle, electron conductivity of the solid electrolyte is another potential mechanism to drive dendrite formation by direct recombination of alkali metal ions and electrons in the solid electrolyte. In $\text{Li}_{6.5}\text{La}_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$ it was experimentally demonstrated to be of significant importance by Han et al. [18]. As this mechanism is only relevant when it occurs prior to critical crack growth, it is interesting at $J < J_{\text{crit}}$.

Mechanical dissipation: Let t_e be the electron transference number of the solid electrolyte, i.e. the quotient of electronic and ionic conductivity. The electronic current field is by definition:

$$\mathbf{j}_e(\mathbf{x}) = t_e \mathbf{j}(\mathbf{x}), \quad \mathbf{x} \in \Omega \quad (35)$$

The consideration of $\mathbf{j}_e$ allows now in principle for deposition anywhere in the solid electrolyte simply by locally recombining electrons and alkali metal ions. However, to deposit an alkali metal atom at oxidation state 0 in the solid electrolyte, the material must be locally destroyed which requires work. In a ceramic without mechanical tension, this work per mole of atoms is simply a material constant, defined as E_f . However, at a global current density $0 < J < J_{\text{crit}}$, the vicinity of the interfacial defect is under mechanical tension below the strength of the ceramic, as depicted in Figure 5.

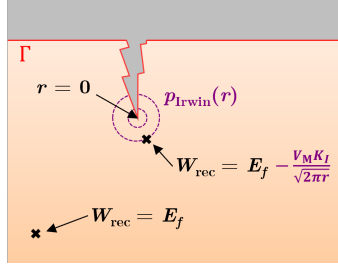


Figure 5: Schematic depiction of the thermodynamical cost reduction of electron-ion-recombination due to the Irwin-field (purple) at the defect tip.

This tension $p_{\text{Irwin}}(r)$ a distance r away from the defect tip (in its direction) follows from Irwin's equation as:

$$p_{\text{Irwin}}(r, J) = \frac{K_I(J)}{\sqrt{2\pi} r} \quad (36)$$

Here $K_I(J) < K_{Ic}$ is the mode I stress intensity factor as a function of J . Let a be the minimal distance electrons need to jump or tunnel from the metallic dendrite tip into the lattice to find an alkali metal ion to recombine with. It is on the length scale of the solid-electrolyte lattice constant and defines the largest lower bound of r for the described mechanism. Consequently, the thermodynamically weakest link for electron-ion recombination in the solid electrolyte is the molar work required to deposit alkali metal atoms at $r = a$ in front of the metal filled crack:

$$W_{\text{rec}}(J) := E_f - \frac{V_M K_I(J)}{\sqrt{2\pi} a} \quad (37)$$

Shifting again to a dynamical picture, the mechanical dissipation associated with such recombination at the defect tip is W_{rec} times the rate of deposition allowed by the electron conduction bottleneck. The largest electron current which is applicable follows from equation 29 as $t_e I_c$. Hence, the mechanical dissipation is:

$$\Delta \dot{W}_{\text{rec}}(J) \lesssim \frac{t_e I_c}{F} W_{\text{rec}} = \frac{\pi t_e J w d}{\mathcal{I}(0)} \left(E_f - \frac{V_M K_I(J)}{\sqrt{2\pi} a} \right) \quad (38)$$

Joule dissipation: By recombining electrons and ions at the dendrite tip, even at $J < J_{\text{crit}}$ some current may be localized. In this case the maximum subcritical localization corresponds to a localized component of the ionic current, equal to the maximum localized electron current. The latter is $\mathbf{j}_e = t_e \mathbf{j}_c$, as defined by equation 18. Hence, the total ionic current field of the maximum subcritical localization around the metal filled defect is:

$$\mathbf{j}_c^{\text{subcrit}}(\mathbf{x}) := (1 - t_e) J \mathbf{e}_z + t_e \mathbf{j}_c(\mathbf{x}) \quad (39)$$

Similar to equation 19, the difference in Joule dissipation when compared to no current localization ($\mathbf{j}_0(\mathbf{x}) = J \mathbf{e}_z$) is calculated as:

$$\Delta \dot{Q}_{\text{rec}} = \frac{1}{\sigma} \int_{\Omega_c} [(1 - t_e) J \mathbf{e}_z + t_e \mathbf{j}_c(\mathbf{x})]^2 dV - \frac{1}{\sigma} \int_{\Omega_c} (J \mathbf{e}_z)^2 dV \approx \frac{2 t_e J}{\sigma} \int_{\Omega_c} [\mathbf{j}_c(\mathbf{x}) \cdot \mathbf{e}_z - J] dV \quad (40)$$

Here, terms quadratic in $t_e \ll 1$ are neglected. The integral is solved in ellipsoidal coordinates using Gauss's theorem as shown in section 4.4, yielding:

$$\Delta \dot{Q}_{\text{rec}} \approx - \frac{2\pi t_e J^2 w d c_{\text{max}}}{3\sigma \mathcal{I}(0)} \quad (41)$$

Criterion for subcritical crack growth: By Onsager's principle, equation 38 and 41 yields a condition for metal deposition inside the solid electrolyte around the dendrite tip and thus subcritical crack growth at $J_{\text{subcrit}} \leq J < J_{\text{crit}}$:

$$\Delta \dot{Q}_{\text{rec}}(J = J_{\text{subcrit}}) + \Delta \dot{W}_{\text{rec}}(J = J_{\text{subcrit}}) = 0 \quad (42)$$

Insertion of equation 38 and 41 and rearranging yields:

$$J_{\text{subcrit}} \approx \frac{3\sigma}{2F c_{\text{max}}} \left(E_f - \frac{V_M K_I}{\sqrt{2\pi a}} \right) \quad (43)$$

K_I is eliminated by insertion of equation 32 at subcritical current density. This is valid, assuming $J = J_{\text{subcrit}}$ already fulfills equation 33. Rearranging yields:

$$J_{\text{subcrit}} = \frac{3\sigma}{2F c_{\text{max}}} \left(E_f - \frac{V_M}{\sqrt{2\pi a}} \cdot \frac{F c_{\text{max}}^{3/2}}{Y \sigma V_M} J_{\text{subcrit}} \right) \Rightarrow J_{\text{subcrit}} \approx \frac{3\sigma E_f}{2F c_{\text{max}}} \cdot \frac{1}{1 + \frac{3\sqrt{c_{\text{max}}}}{2Y\sqrt{2\pi a}}} \quad (44)$$

Since $a \ll c_{\text{max}}$, the second term of the denominator dominates, yielding approximately:

$$J_{\text{subcrit}} \approx \frac{Y \sigma E_f \sqrt{2\pi a}}{F c_{\text{max}}^{3/2}} = \frac{Y \sigma V_M K_{Isc}}{F c_{\text{max}}^{3/2}} \quad \text{with} \quad K_{Isc} := \frac{E_f \sqrt{2\pi a}}{V_M} \quad (45)$$

Here, K_{Isc} is defined as the stress-corrosion-cracking threshold of the stress intensity factor. The onset of subcritical crack growth inherits the same $c^{-3/2}$ dependence as equation 32. J_{subcrit} increases above J_{crit} , and thus becomes irrelevant, as recombination becomes locally expansive (large E_f) or if the smallest electron jump distance a becomes large. Both are material properties. For sufficiently small E_f and a it predicts electrochemical stress-corrosion-cracking and dendrite propagation at $J_{\text{subcrit}} \leq J < J_{\text{crit}}$.

Subcritical failure time: Consider $J_{\text{subcrit}} \leq J < J_{\text{crit}}$. Alkali metal is deposited at the tip of the dendrite, slowly lengthening it. To estimate the growth time to failure, consider equation 29. The volume injection rate $\dot{V}_{\text{subcrit}}$ at the dendrite is roughly:

$$\dot{V}_{\text{subcrit}}(J) \approx \frac{t_e I_c V_M}{F} = \frac{\pi t_e w d V_M}{F \mathcal{I}(0)} J \approx \frac{\pi t_e c_{\text{max}}^2 V_M}{F \ln(c_{\text{max}}/w)} J \quad (46)$$

Here, the approximation of a thin prolate ellipsoid was applied (equation 34). Given the cross-section of the dendrite scales like πw^2 , the crack grows with:

$$\dot{c}_{\text{max}}(J) \approx \frac{\dot{V}_{\text{subcrit}}}{\pi w^2} \approx \frac{t_e c_{\text{max}}^2 V_M}{F \ln(c_{\text{max}}/w) w^2} J \quad (47)$$

With $\ln(c/w) \sim 1$, an initial length c_{max}^0 and a failure length $c_{\text{max}}^{\text{fail}}$ at which the given value of J fulfill equation 32 and the dendrite will grow by critical fracture of the solid electrolyte, it follows for the fail time τ_{fail} :

$$\int_{c_{\text{max}}^0}^{c_{\text{max}}^{\text{fail}}} \frac{dc}{c^2} \approx \frac{t_e V_M J}{F w^2} \int_0^{\tau_{\text{fail}}} dt \Rightarrow \frac{1}{c_{\text{max}}^0} - \frac{1}{c_{\text{max}}^{\text{fail}}} \approx \frac{t_e V_M J}{F w^2} \tau_{\text{fail}} \quad (48)$$

For $c_{\text{max}}^{\text{fail}} \gg c_{\text{max}}^0$ thus:

$$\tau_{\text{fail}} \approx \frac{F w^2}{t_e V_M J c_{\text{max}}^0} \quad (49)$$

As it turns out, even if E_f (e.g. at grain boundaries) is small and J_{subcrit} is reached during operation of the SSB, subcritical failure can be suppressed by $t_e \approx 0$.

2.8 Weibull-statistics of dendrite formation

Equation 32 and 45 demonstrate a dependence of J_{crit} and J_{subcrit} on the largest defect c_{max} at the interface - provided the defect is sufficiently thin to constrain alkali metal creep inside it (equation 33). In 1939 Weibull applied the same weakest link statistics argument to Griffith's equation for mechanical strength to describe

scattering between ceramic samples. Weibull's approach can be extended to dendrite growth statistics, as shown in the following. The proof is carries out for J_{crit} . It is equivalent for J_{subcrit} .

Largest defect in a sample: Assuming the N defects on Γ are statistically independent, their Pareto distribution (equation 1) implies the cumulative distribution function of the largest defect in a given sample:

$$P(c_{\text{max}} < c) = [P(C < c)]^N = \left[1 - \left(\frac{c_{\text{min}}}{c}\right)^k\right]^N \quad \text{for} \quad c \geq c_{\text{min}}. \quad (50)$$

Cumulative distribution function of J_{crit} : For readability, the constant κ is introduced to rewrite equation 32 as

$$J_{\text{crit}} =: \kappa c_{\text{max}}^{-3/2}. \quad (51)$$

Since J_{crit} decreases as c_{max} increases, there exists some value $\tilde{c}$ such that

$$J_{\text{crit}} \leq \tilde{J}_{\text{crit}} := \kappa \tilde{c}^{-3/2} \quad \Leftrightarrow \quad \tilde{c} = \left(\frac{\kappa}{\tilde{J}_{\text{crit}}}\right)^{2/3} \leq c_{\text{max}}. \quad (52)$$

Consequently, the scattering of J_{crit} can be expressed by the scattering of $\tilde{c}$ as

$$P(J_{\text{crit}} \leq \tilde{J}_{\text{crit}}) = P(c_{\text{max}} \geq \tilde{c}) = 1 - P(c_{\text{max}} < \tilde{c}) = 1 - \left[1 - \left(\frac{c_{\text{min}}}{\tilde{c}}\right)^k\right]^N. \quad (53)$$

lower tail approximation: The technologically relevant case of determining the statistics for $\tilde{c} \gg c_{\text{min}}$ yields $(c_{\text{min}}/\tilde{c})^k \ll 1$. Therefore, and due to $N \gg 1$ equation 53 can be approximated as

$$P(J_{\text{crit}} \leq \tilde{J}_{\text{crit}}) = 1 - \left[1 - \left(\frac{c_{\text{min}}}{\tilde{c}}\right)^k\right]^N \approx 1 - \exp\left[-N \left(\frac{c_{\text{min}}}{\tilde{c}}\right)^k\right]. \quad (54)$$

Partitioning the interfacial area A_Γ into N average correlation areas A_c per relevant interfacial defect, and reinserion of $\tilde{c}$ and κ , yields:

$$P(J_{\text{crit}} \leq \tilde{J}_{\text{crit}}) \approx 1 - \exp\left[-\frac{A_\Gamma}{A_c} \left(\frac{F c_{\text{min}}^{3/2} \tilde{J}_{\text{crit}}}{Y \sigma V_M K_{Ic}}\right)^{2k/3}\right]. \quad (55)$$

With Weibull-modulus m and scale parameter J_0 , where

$$m = \frac{2k}{3} \quad \text{and} \quad J_0 = \frac{Y V_M K_{Ic} \sigma}{F c_{\text{min}}^{3/2}}, \quad (56)$$

this is a Weibull-distribution:

$$\boxed{P(J_{\text{crit}} \leq \tilde{J}_{\text{crit}}) \approx 1 - \exp\left[-\frac{A_\Gamma}{A_c} \left(\frac{\tilde{J}_{\text{crit}}}{J_0}\right)^m\right]} \quad (57)$$

In a similar manor, a shifted Weibull-distribution can be derived, by fixing an upper cutoff C_{max} for the Pareto distribution of defect lengths in equation 1, i.e. a largest allowable defect size C_{max} that can be present in any sample. This then shifts the Weibull-distribution by a threshold

$$J_u = \frac{Y \sigma V_M K_{Ic}}{F C_{\text{max}}^{3/2}} \quad (58)$$

below which no dendrites can be observed in any sample:

$$\boxed{P(J_{\text{crit}} \leq \tilde{J}_{\text{crit}}) \approx 1 - \exp\left[-\frac{A_\Gamma}{A_c} \left(\frac{\tilde{J}_{\text{crit}} - J_u}{J_0}\right)^m\right]} \quad (59)$$

2.9 Numerical example and plausibility discussion

Experimentally, values of $J_{\text{crit}} \sim 1 \text{ mA/cm}^2$ have been measured for ceramic sodium solid electrolytes of the composition $\text{Na}_5\text{SmSi}_4\text{O}_{12}$ [12].

Numerical assumptions: FESEM-images by Anton, Schilm et al. [19, 20] exhibit thin micro-cracks between grains with $c \sim 10 \mu\text{m}$ and $w \sim 50 \text{ nm}$. A value of $c_{\text{max}} \sim 30 \mu\text{m}$ is plausible. For these values, $\mathcal{I}(0) \approx 0.001$ is found by numerical integration of equation 26. The geometry parameter is assumed to be $Y \approx 3/2$. The ionic conductivity of the materials in the range of $\sigma \sim 2 \text{ mS/cm}$ [20, 19]. Constants are $V_{\text{Na}} = 23.7 \text{ cm}^3/\text{mol}$ and $F = 96485 \text{ C/mol}$. The fracture toughness is assumed to be a typical oxide ceramic value of $K_{Ic} \approx 1 \frac{\text{MPa}}{\sqrt{\text{m}}}$. Based on measurements by Wang et al. [21], $G = 0.928 \text{ MPas}^n$ and $n = 0.164$ is assumed to model the power-law creep of sodium at room temperature.

Necessary condition: Insertion of the numerical values into equation 33 yields a necessary global current density of $J_{\text{crit}} \gtrsim 0.001 \text{ mA/cm}^2$ for the micro-crack defects to be relevant for dendrite formation. As this is far below the experimentally observed range, it can be reasonably concluded that this necessary condition has been satisfied.

Critical current density: Since the defects are relevant for dendrite formation, the numerical assumptions can now be inserted into equation 32, yielding $J_{\text{crit}} \sim 50 \text{ mA/cm}^2$. This value is one to two orders of magnitude higher than the experimentally observed range. This is however not surprising, as the model treated K_{Ic} as a material constant. This assumption is almost certainly wrong and has been topic of heated discussion in the last years. While atomistic simulations by Zhang et al. [22] suggest mechanisms by which K_{Ic} increases at the tip of a dendrite, due to local deprivation of lithium ions, other mechanisms predict the opposite. Yu and Siegel [23] demonstrate via molecular dynamics simulations a reduction in shear modulus at grain boundaries compared to solid electrolyte crystals. A lowering of shear modulus directly implies lowering of K_{Ic} . Phase changes at dendrite tips are another possible mechanism to effectively decrease K_{Ic} as recently suggested by Fincher et al. [24]. Also, K_{Ic} and thus J_{crit} will be inherently decreased for significant work of adhesion between solid electrolyte and alkali metal electrode (equation 13).

Subcritical dendrite growth: Estimating numerical values of J_{subcrit} in equation 45 is difficult without simulations to determine E_f . However, assuming $a \approx 0.5 \text{ nm}$ and $E_f \approx 5 \text{ eV}$ as sensible order of magnitudes together with the same values as above yields $J_{\text{subcrit}} \sim 50 \text{ mA/cm}^2$. Hence, J_{subcrit} and J_{crit} are on the same order of magnitude. Hence subcritical dendrite growth might not be important. However, if E_f is significantly smaller at grain boundaries compared to the lattice, which is to be expected for most materials, J_{subcrit} might be significantly smaller than J_{crit} . With the assumptions made above, $t_e \sim 10^{-6}$ and the experimentally observed current magnitude of $J \sim 1 \text{ mA/cm}^2$, the relevant timescale of stress-corrosion-cracking follows from equation 49 as $\tau_{\text{fail}} \sim 10 \text{ hours}$. For the calculated value of $J \sim 50 \text{ mA/cm}^2$ correspondingly $\tau_{\text{fail}} \sim 10 \text{ minutes}$. Should J_{subcrit} turn out to be small enough, the time scale of subcritical dendrite growth would be technologically relevant!

Statistical scattering: Critical current density measurements on $\text{Na}_5\text{SmSi}_4\text{O}_{12}$ were best described by a Weibull-distribution with a Weibull-modulus of $m = 1.1$ [12]. When again considering the microstructure of the solid electrolyte samples (FESEM images in [19, 20]), it becomes clear that one of the thin grain boundary cracks which was inserted into the presented equations to calculate J_{crit} will likely also be the largest defect in any mechanical strength test (such like a ball on three ball experiment). Consequently, the Pareto-exponent k in equation 1 will be approximately the same for both electrochemical and mechanical failure. The mechanical Weibull-modulus is known to be $m_{\text{mech}} = 2k$. According to equation 56, this is three times larger than the electrochemical Weibull-modulus. Therefore, a value of $m_{\text{mech}} = 3.3$ is expected for the same solid electrolyte. While detailed tests on the exact material are still pending, Wagner measured values in the rage of $m_{\text{mech}} = 3 - 9$ on very similar compositions in her doctoral thesis [25]. Hence, the expected value is within the measured range.

3 Conclusion

A quantitative model of dendrite penetration through solid electrolytes was derived by comparing Joule dissipation during the transport of the ionic current through the solid electrolyte with the mechanical dissipation during interfacial metal deposition. By the thermodynamical argument of minimal total dissipation, and by approximating the shape of interfacial defects as ellipsoids, two conditions for dendrite growth were derived:

$$J_{\text{crit}} \approx \frac{Y \sigma V_{\text{M}} K_{\text{Ic}}}{F c_{\text{max}}^{3/2}}, \quad J_{\text{subcrit}} \approx \frac{Y \sigma E_f \sqrt{2\pi a}}{F c_{\text{max}}^{3/2}} \quad (60)$$

The first expression J_{crit} describes the smallest current density at which a dendrite will propagate through the solid electrolyte via critical crack propagation. It depends on the length of the longest interfacial defect c_{max} (assuming this defect is sufficiently thin) and on material constants (ionic conductivity σ , fracture toughness K_{Ic} of the solid electrolyte in contact with the alkali metal, molar volume V_{M} of the alkali metal, the geometry parameter $Y \sim 3/2$ and the Faraday constant F).

The second expression J_{subcrit} has the same algebraic form as J_{crit} and describes the current density above which a dendrite can grow subcritically from the interface to increase c_{max} until $J = J_{\text{crit}}$ is reached. The underlying mechanism is the recombination of alkali metal ions and residual conduction electrons inside the solid electrolyte. It depends on the work E_f required to reduce an ion inside the lattice or grain boundary without mechanical tension, the minimal jump distance a of ions at the dendrite tip and the electron transference number t_e . The latter determines the timescale $\tau_{\text{fail}} \propto t_e^{-1}$ until critical failure.

The two formulas provide clear guidelines for designing solid electrolytes resistant to dendrite penetration. Microstructures containing long and thin defects, such as grain-boundary micro-cracks, drive J_{crit} to undesirably low values. Notable progress toward suppressing such defects has been reported by Liu et al. [26]. Other strategies include an increase of the ionic conductivity σ and the fracture toughness K_{Ic} . The mechanisms by which the latter decreases at dendrite tips remain poorly understood [24, 23, 22]. Since the model predicts the interfacial defect responsible for the initiation of dendrite growth to be filled with alkali metal, K_{Ic} decreases as the work of adhesion between electrode and solid electrolyte increases. This result might be counter intuitive and poses a design dilemma with respect to other challenges of the same interface (such as interfacial void formation during cell discharge [27]). To avoid subcritical crack growth, E_f must be sufficiently large. How this quantity depends on crystal structure and whether it is reduced significantly at grain boundaries must be explored in future studies to evaluate the relevance of the proposed mechanism. Should E_f be undesirably small, subcritical stress-corrosion-cracking can be suppressed by reducing the electron conductivity of the solid electrolyte.

The defect dependence of both J_{crit} and J_{subcrit} rationalizes the weak-link penetration behavior documented in [12] and explains the pronounced scatter of J_{crit} between samples of similar preparation. The model postulates this scattering to be Weibull-distributed. The theoretical Weibull-modulus should approximately be one third of the Weibull-modulus found in mechanical strength tests of the solid electrolyte. Such Weibull-distributed scattering at low Weibull-modulus has previously been experimentally observed [12] and can now be theoretically understood. Following these elaborations, the statistical thinking commonly applied to the mechanical properties of ceramics should be expanded to the critical current densities of ceramic solid electrolytes.

4 Supplementary derivations

4.1 Gradient and normal derivative of Φ_c

To derive the electric field around the ellipsoid, the gradient of equation 25 must be calculated:

$$-\nabla\Phi_c(\lambda, z) = \mathbf{E}(\lambda, z) = \nabla \left\{ E_0 z \left[1 - \frac{\mathcal{I}(\lambda)}{\mathcal{I}(0)} \right] \right\} \quad (61)$$

Applying the product and chain rule, this yields:

$$-\nabla\Phi_c(\lambda, z) = E_0 \nabla z \left[1 - \frac{\mathcal{I}(\lambda)}{\mathcal{I}(0)} \right] + E_0 z \left[1 - \frac{\mathcal{I}'(\lambda)}{\mathcal{I}(0)} \right] \nabla \lambda \quad (62)$$

Finding the gradient $\nabla z = \mathbf{e}_z$ is trivial. Calculating $\mathcal{I}'(\lambda)$ from equation 26 is also trivial, as differentiation simply cancels integration. Finding the gradient $\nabla \lambda$ is an implicit problem of the defining equation of the ellipsoids surface:

$$\frac{x^2}{w^2 + \lambda} + \frac{y^2}{d^2 + \lambda} + \frac{z^2}{c^2 + \lambda} = 1 \quad (63)$$

Differentiating this equation with respect to x yields:

$$-\frac{x^2}{(w^2 + \lambda)^2} \frac{\partial \lambda}{\partial x} + \frac{2x}{w^2 + \lambda} - \frac{y^2}{(d^2 + \lambda)^2} \frac{\partial \lambda}{\partial x} - \frac{z^2}{(c^2 + \lambda)^2} \frac{\partial \lambda}{\partial x} = 0 \quad (64)$$

Rearranging yields $\frac{\partial \lambda}{\partial x}$. Likewise, differentiation with respect to y and z yields $\frac{\partial \lambda}{\partial y}$ and $\frac{\partial \lambda}{\partial z}$. When defining (for readability) the shorthand

$$T(\lambda) = \frac{x^2}{(w^2 + \lambda)^2} + \frac{y^2}{(d^2 + \lambda)^2} + \frac{z^2}{(c^2 + \lambda)^2}, \quad (65)$$

this yields:

$$\nabla \lambda = \frac{2}{T(\lambda)} \left(\frac{x}{w^2 + \lambda} \mathbf{e}_x + \frac{y}{d^2 + \lambda} \mathbf{e}_y + \frac{z}{c^2 + \lambda} \mathbf{e}_z \right) \quad (66)$$

Insertion of ∇z , $\nabla \lambda$ and $\mathcal{I}'(\lambda)$ (equation 26) into equation 62 finally yields:

$$-\nabla\Phi_c(\lambda, z) = E_0 \left[1 - \frac{\mathcal{I}(\lambda)}{\mathcal{I}(0)} \right] \mathbf{e}_z + \frac{E_0 z w d c}{\mathcal{I}(0) T(\lambda) (c^2 + \lambda) R(\lambda)} \left(\frac{x}{w^2 + \lambda} \mathbf{e}_x + \frac{y}{d^2 + \lambda} \mathbf{e}_y + \frac{z}{c^2 + \lambda} \mathbf{e}_z \right) \quad (67)$$

By definition, $\lambda = 0$ defines the surface $\Gamma_c \cup \Gamma'_c$ of the ellipsoid. Here, and thus on the defect's surface Γ_c , the gradient should become purely normal. Insertion of $\lambda = 0$ and $R(0)$ (equation 24) into equation 67 yields indeed:

$$-\nabla\Phi_c|_{\Gamma_c} = \frac{E_0 z}{\mathcal{I}(0) c^2 T(0)} \left(\frac{x}{w^2} \mathbf{e}_x + \frac{y}{d^2} \mathbf{e}_y + \frac{z}{c^2} \mathbf{e}_z \right) = \frac{E_0 z}{\mathcal{I}(0) c^2 \sqrt{T(0)}} \mathbf{n} \quad (68)$$

4.2 Derivation of $\Delta\dot{Q}_{\max}$

The aim is to solve the integral of equation 19 to obtain equation 27. To ease the derivation, rewrite equation 19 as:

$$\Delta\dot{Q}^* := \Delta\dot{Q}_{\max}(J) - \sigma E_0^2 V_c = \sigma \int_{\Omega_c} (|\nabla\Phi_c|^2 - E_0^2) dV \quad (69)$$

4.2.1 Volume integration to surface integration via Gauss's theorem

Begin by rewriting the potential Φ_c as:

$$\Phi_c = \Phi_0 + \Phi^* \quad (70)$$

Φ_0 is the limit of the electrical potential far away from the defect and thus the potential for the non-localized case of equation 18. Φ^* is the potential perturbation in the vicinity of the defect. Consequently, Φ^* must decay to zero at infinity and satisfy the Laplace equation, i.e. $\nabla^2 \Phi^* = 0$. Insertion into equation 19 yields

$$\Delta \dot{Q}^* = \sigma \int_{\Omega_c} (|\nabla \Phi_0 + \nabla \Phi^*|^2 - E_0^2) dV = \sigma \int_{\Omega_c} (2\nabla \Phi_0 \nabla \Phi^* + |\nabla \Phi^*|^2) dV. \quad (71)$$

For simplicity, integrate the mixed and the quadratic term independently.

Mixed term: Using the product rule and $\nabla^2 \Phi^* = 0$ in Ω_c , the integration of the mixed term yields:

$$\int_{\Omega_c} 2\nabla \Phi_0 \nabla \Phi^* dV = 2 \int_{\Omega_c} [\nabla \cdot (\Phi_0 \nabla \Phi^*) - \underbrace{\Phi_0 \nabla^2 \Phi^*}_{=0}] dV = 2 \int_{\Omega_c} \nabla \cdot (\Phi_0 \nabla \Phi^*) dV. \quad (72)$$

This expression can be rewritten using Gauss's theorem using the outward pointing unit normal vector $\mathbf{n}$ of the surface $\partial\Omega_c$ of Ω_c :

$$2 \int_{\Omega_c} \nabla \cdot (\Phi_0 \nabla \Phi^*) dV = 2 \oint_{\partial\Omega_c} \Phi_0 \nabla \Phi^* \cdot \mathbf{n} dA = 2 \oint_{\partial\Omega_c} \Phi_0 \frac{\partial \Phi^*}{\partial n} dA. \quad (73)$$

Following the illustration in Figure 4, $\partial\Omega_c$ can now be separated into the ellipsoids surface $\Gamma_c \cup \Gamma'_c$ and a spheroid surface $\partial\Omega_{c,\infty}$ of infinite radius, hence:

$$2 \oint_{\partial\Omega_c} \Phi_0 \frac{\partial \Phi^*}{\partial n} dA = 2 \oint_{\Gamma_c \cup \Gamma'_c} \Phi_0 \frac{\partial \Phi^*}{\partial n} dA + 2 \oint_{\partial\Omega_{c,\infty}} \Phi_0 \frac{\partial \Phi^*}{\partial n} dA. \quad (74)$$

The integration over $\partial\Omega_{c,\infty}$ vanishes, since $\Phi^* \sim r^{-2}$ far away from the defect (proof my multipole expansion) and thus $\frac{\partial \Phi^*}{\partial n} \sim r^{-3}$. Therefore, the mixed term of equation 71 reduces to an integral over the ellipsoids surface:

$$\int_{\Omega_c} 2\nabla \Phi_0 \nabla \Phi^* dV = 2 \oint_{\Gamma_c \cup \Gamma'_c} \Phi_0 \frac{\partial \Phi^*}{\partial n} dA. \quad (75)$$

Quadratic term: To calculate the quadratic term of equation 71, consider $\nabla^2 \Phi^* = 0$ and use the product rule once again:

$$\int_{\Omega_c} |\nabla \Phi^*|^2 dV = \int_{\Omega_c} [\nabla \cdot (\Phi^* \nabla \Phi^*) - \underbrace{\Phi^* \nabla^2 \Phi^*}_{=0}] dV = \int_{\Omega_c} \nabla \cdot (\Phi^* \nabla \Phi^*) dV. \quad (76)$$

Analogue to equation 73 and 74, Gauss's theorem is again applied and surface integration over $\partial\Omega_{c,\infty}$ vanishes:

$$\int_{\Omega_c} |\nabla \Phi^*|^2 dV = \oint_{\Gamma_c \cup \Gamma'_c} \Phi^* \frac{\partial \Phi^*}{\partial n} dA. \quad (77)$$

Combining both terms: Now equation 75 and 77 are inserted into equation 71, yielding:

$$\Delta \dot{Q}^* = 2\sigma \oint_{\Gamma_c \cup \Gamma'_c} \Phi_0 \frac{\partial \Phi^*}{\partial n} dA + \sigma \oint_{\Gamma_c \cup \Gamma'_c} \Phi^* \frac{\partial \Phi^*}{\partial n} dA. \quad (78)$$

Now consider the boundary condition, the surface of the metal filled defect (i.e. the prolate ellipsoid) and the interface between solid electrolyte and electrode are of constant potential. Since the value of this interfacial potential is well defined up to an additive constant, choose arbitrarily $\Phi_c|_{\Gamma_c \cup \Gamma'_c} = 0$. Hence equation 70 yields:

$$\Phi^*|_{\Gamma_c \cup \Gamma'_c} = -\Phi_0|_{\Gamma_c \cup \Gamma'_c}. \quad (79)$$

Insertion into equation 78 simplifies to:

$$\Delta\dot{Q}^* = \sigma \oint_{\Gamma_c \cup \Gamma'_c} \Phi_0 \frac{\partial \Phi^*}{\partial n} dA \quad (80)$$

Substituting Φ^* via equation 70 yields:

$$\Delta\dot{Q}^* = \sigma \oint_{\Gamma_c \cup \Gamma'_c} \Phi_0 \frac{\partial \Phi_c}{\partial n} dA - \sigma \oint_{\Gamma_c \cup \Gamma'_c} \Phi_0 \frac{\partial \Phi_0}{\partial n} dA \quad (81)$$

By the ellipsoid equation, the ellipsoid's surface is in cartesian coordinates an implicit function:

$$f(x, y, z) = \frac{x^2}{w^2} + \frac{y^2}{d^2} + \frac{z^2}{c^2} - 1 = 0 \quad (82)$$

By definition and when using the definition of $T(0)$ (equation 65), the normal unit vector on this surface in cartesian coordinates is thus:

$$\mathbf{n} = \frac{\nabla f}{|\nabla f|} = \frac{1}{\sqrt{T(0)}} \left(\frac{x}{w^2} \mathbf{e}_x + \frac{y}{d^2} \mathbf{e}_y + \frac{z}{c^2} \mathbf{e}_z \right) \quad (83)$$

Use this result, to realize that:

$$\left. \frac{\partial \Phi_0}{\partial n} \right|_{\Gamma_c \cup \Gamma'_c} = \nabla \Phi_0 \cdot \mathbf{n} \Big|_{\Gamma_c \cup \Gamma'_c} = -E_0 \mathbf{e}_z \cdot \mathbf{n} = -\frac{E_0 z}{c^2 \sqrt{T(0)}} \quad (84)$$

Equation 68 and 84 are now inserted into equation 81. Together with $\Phi_0 = -E_0 z$, this yields:

$$\Delta\dot{Q}^* = \frac{\sigma E_0^2}{c^2} \left(\frac{1}{\mathcal{I}(0)} - 1 \right) \oint_{\Gamma_c \cup \Gamma'_c} \frac{z^2}{\sqrt{T(0)}} dA \quad (85)$$

4.2.2 Solving the surface integral

Reparametrization: To solve the integral, reparametrize the ellipsoid's surface in angular coordinates:

$$x = w \sin \theta \cos \varphi, \quad y = d \sin \theta \sin \varphi, \quad z = c \cos \theta, \quad \theta \in [0, \pi], \quad \varphi \in [0, 2\pi]. \quad (86)$$

Surface element: Recall that for a surface which is parametrized by $\mathbf{r}(\theta, \varphi)$, the surface element is given by

$$dA = |\mathbf{r}_\theta \times \mathbf{r}_\varphi| d\theta d\varphi. \quad (87)$$

The two tangent vectors on the ellipsoid follow from equation 86 by component-wise differentiation as:

$$\mathbf{r}_\theta = \begin{pmatrix} w \cos \theta \cos \varphi \\ d \cos \theta \sin \varphi \\ -c \sin \theta \end{pmatrix}, \quad \mathbf{r}_\varphi = \begin{pmatrix} -w \sin \theta \sin \varphi \\ d \sin \theta \cos \varphi \\ 0 \end{pmatrix} \quad (88)$$

By definition, $\mathbf{r}_\theta$ and $\mathbf{r}_\varphi$ are tangent to the surface normal ∇f (equation 82 and 83). Consequently, $\mathbf{r}_\theta \times \mathbf{r}_\varphi$ — which is also perpendicular to both tangent vectors — must be parallel to ∇f . Hence

$$\mathbf{r}_\theta \times \mathbf{r}_\varphi = \xi(\theta, \varphi) \nabla f \quad (89)$$

for some scalar ξ . To find ξ , calculate the z -component of the cross product directly:

$$(\mathbf{r}_\theta \times \mathbf{r}_\varphi)_z = (r_\theta)_x (r_\varphi)_y - (r_\theta)_y (r_\varphi)_x = w \cos \theta \cos \varphi \cdot d \sin \theta \cos \varphi + d \cos \theta \sin \varphi \cdot w \sin \theta \sin \varphi = wd \sin \theta \cos \theta \quad (90)$$

By equation 83, the z -component of $\nabla f = 2(x/w^2, y/d^2, z/c^2)^T$ evaluated on the surface is

$$(\nabla f)_z = \frac{2z}{c^2} = \frac{2c \cos \theta}{c^2} = \frac{2 \cos \theta}{c}. \quad (91)$$

Equating $(\mathbf{r}_\theta \times \mathbf{r}_\varphi)_z = \xi(\nabla f)_z$ yields:

$$w d \sin \theta \cos \theta = \xi \frac{2 \cos \theta}{c} \quad \implies \quad \xi = \frac{w d c \sin \theta}{2} \quad (92)$$

Using $|\nabla f| = 2\sqrt{T(0)}$ from equation 83 this yields:

$$|\mathbf{r}_\theta \times \mathbf{r}_\varphi| = \xi |\nabla f| = \frac{w d c \sin \theta}{2} \cdot 2\sqrt{T(0)} = w d c \sqrt{T(0)} \sin \theta \quad (93)$$

Hence, the surface element follows by insertion in equation 87 as:

$$dA = w d c \sqrt{T(0)} \sin \theta d\theta d\varphi. \quad (94)$$

Computation: The parametrization of z (equation 86) and the surface element dA (equation 94) are inserted into equation 85. This yields:

$$\oint_S \frac{z^2}{\sqrt{T(0)}} dA = w d c \int_0^{2\pi} d\varphi \int_0^\pi c^2 \cos^2 \theta \sin \theta d\theta = w d c^3 \cdot 2\pi \cdot \frac{2}{3} = \frac{4\pi w d c^3}{3} \quad (95)$$

Insertion into equation 85 gives:

$$\Delta \dot{Q}^* = \left(\frac{1}{\mathcal{I}(0)} - 1 \right) \underbrace{\frac{4\pi w d c}{3}}_{V_c} \sigma E_0^2 \quad (96)$$

Notice, that the volume V_c of the ellipsoid appears! Inserting this expression back into the original equation 69 yields:

$$\Delta \dot{Q}_{\max} = \frac{4\pi \sigma E_0^2 w d c}{3 \mathcal{I}(0)} \quad (97)$$

Division by 2, since only the dissipation reduction in the lower half-space is of interest (by the method of image charge) derives the desired expression of equation 19. $\square$

4.3 Derivation of I_c

The aim is to solve the integral of equation 28 to derive equation 29. Into

$$I_c = -\sigma \int_{\Gamma_c} \nabla \Phi_c \cdot \mathbf{n} dA, \quad (98)$$

equation 67 is inserted for $\nabla \Phi_c$, yielding:

$$I_c = \frac{\sigma E_0}{\mathcal{I}(0) c^2} \int_{\Gamma_c} \frac{z}{\sqrt{T(0)}} dA \quad (99)$$

The integral is again parametrized in angular coordinates via equation 86 and 94:

$$I_c = \frac{\sigma E_0 w d}{\mathcal{I}(0)} \int_0^{2\pi} d\varphi \int_0^{\pi/2} \cos \theta \sin \theta d\theta = -\frac{\pi \sigma E_0 w d}{\mathcal{I}(0)} [\cos^2(\theta)]_0^{\pi/2} = \frac{\pi \sigma E_0 w d}{\mathcal{I}(0)} \quad (100)$$

This is the desired expression of equation 29. $\square$

4.4 Derivation of $\Delta\dot{Q}_{\text{rec}}$

Equation 41 must be derived. From equation 40, $\mathbf{j}_c = -\sigma\nabla\Phi_c$ and $J = \sigma E_0$ follows:

$$\Delta\dot{Q}_{\text{rec}} = \frac{2t_e J}{\sigma} \int_{\Omega_c} [\mathbf{j}_c(\mathbf{x}) \cdot \mathbf{e}_z - J] \, dV = -2t_e J \int_{\Omega_c} [\partial_z \Phi_c + E_0] \, dV = -2t_e J \int_{\Omega_c} \partial_z \Phi^* \, dV \quad (101)$$

Using Gauss's theorem, $\nabla^2 \Phi^* = 0$ in Ω_c and vanishing of Φ^* at infinity:

$$\int_{\Omega_c} \partial_z \Phi^* \, dV = \int_{\Omega_c} \nabla \cdot (\Phi^* \mathbf{e}_z) \, dV = \oint_{\partial\Omega_c} \Phi^* (\mathbf{e}_z \cdot \mathbf{n}) \, dA = \oint_{\Gamma_c \cup \Gamma'_c} \Phi^* (\mathbf{e}_z \cdot \mathbf{n}) \, dA \quad (102)$$

On $\Gamma_c \cup \Gamma'_c$ the boundary condition $\Phi_c = 0$ holds, thus $\Phi^* = -\Phi_0 = E_0 z$:

$$\oint_{\Gamma_c \cup \Gamma'_c} \Phi^* (\mathbf{e}_z \cdot \mathbf{n}) \, dA = E_0 \oint_{\Gamma_c \cup \Gamma'_c} z (\mathbf{e}_z \cdot \mathbf{n}) \, dA = \frac{E_0}{c^2} \oint_{\Gamma_c \cup \Gamma'_c} \frac{z^2}{\sqrt{T(0)}} \, dA \quad (103)$$

This is the surface integral of equation 85 which was solved in section 4.2.2. This yields the desired expression of equation 41. $\square$

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